

PRINCIPAL COMPONENTS ANALYSIS AND RESYNTHESIS OF SPECTRAL CUES TO PERCEIVED DIRECTION

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Head-related transfer functions (HRTF's) were measured using Time-Delay Spectrometry (TDS) for two subjects at 36 sound-source azimuth angles on the horizontal plane (at ear level). The variation in magnitude of these HRTF's was observed within 24 spectral bands modeling the critical bands of the peripheral auditory system (Petersen, 1980). These transformed HRTF data were submitted to principal components analysis (PCA) in order to determine a simplified representation of the spectral cues to perceived azimuth. This application of PCA, sometimes referred to as spectral PCA, is similar to Stautner's (1983) technique for encoding time-varying spectra, but encodes spatial variation in spectra. The first few principal components obtained from the analysis of the horizontal plane data distinguished directionality of HRTF's for the following sets of azimuth angles: 1) ipsilateral versus contralateral, 2) front versus rear, 3) central versus lateral, and 4) odd quadrants versus even quadrants (The odd quadrants of azimuth angles are ipsilateral-front and contralateral-rear). In each case, the obtained scores could be fit by smooth curves defining smooth paths through the low-dimensional principal components space, and providing high-quality interpolation for resynthesized spectra. Resynthesis achieved through the inversion of the principal components rotation matrix can be driven from synthetic principal component scores as well, yielding novel sets of spectra. Furthermore, idealized synthetic scores have yielded sets of resynthesized HRTF's that have served to refine idealized directional transfer functions (DTF's) simulating spectral cues to directional hearing (Kendall & Martens, 1984).

1. Introduction

The perceived location of sound sources in space has two distinct aspects - the apparent direction of the sound source and the apparent distance of the sound source. This paper is concerned only with the acoustical information that provides cues to the apparent direction of sounds in space, and is further constrained to consider only the perception of the direction of sound sources located on the horizontal plane - the plane defined by the left/right and the front/rear axes (all possible directions at ear level). The direction of a sound source on the horizontal plane is measured by its azimuth angle, starting at 0 degrees for a sound source located directly ahead of the listener and increasing in a clockwise direction. The acoustical information allowing the listener to identify the azimuth angle of a sound source is captured by the binaural head-related transfer function (HRTF) for that azimuth angle. Though it may be that the acoustical information present at one ear is sufficient for accurate directional judgments (Butler & Flannery, 1982), there is additional information carried in the pair of transfer functions for each azimuth angle that could provide a binaural contribution to normal directional perception. Therefore, the exploration of contralateral-ear involvement in the encoding of sound-source direction has not been excluded from the present investigation.

Though there has been much research on the role that HRTF's play in directional hearing (see Blauert [1982] for a comprehensive review), no well-integrated account of the role played by HRTF's has been formulated (cf. Searle, Braida, Davis, & Colburn, 1976). In the interest of developing such a global account, a two-stage research project has been initiated to examine and determine what information is present in the HRTF's that is used in human directional perception. The first stage is strictly exploratory, examining how HRTF's vary with changing sound-source direction. The second stage is confirmatory, attempting to determine whether the information revealed in the first stage of research can be used to predict directional perception. This paper reports preliminary findings of the exploratory first stage of our research using principal components analysis (PCA). This technique of multivariate analysis has been used as a data reduction procedure for the overall spectral variation in HRTF's that is associated with changing direction on the horizontal plane.

The domain in which HRTF's encode directional information is still controversial after roughly twenty years of research. Batteau (1967) attempted to identify individual pinna reflections responsible for judgements of azimuth and elevation. Blauert (1969) hypothesized that pinna cues are evaluated by the auditory system in terms of "directional bands" that map spectral patterns

into perceived direction. Though it has been suggested that time-domain envelopes might carry directional information (Hiranaka and Yamasaki, 1983), the more popular view is currently that spectral cues are most significant. Of course, sound reflected off the convolutions of the pinna, shoulders and the upper torso will produce changes in the spectrum of an incident sound signal, so the time vs. frequency domain descriptions are not necessarily contradictory (Wright, Hebrank and Wilson, 1974). In the final analysis, close examination makes it clear that an accurate account of the directionally-dependent changes in HRTF's must encompass multiple perspectives on the phenomenon:

"The acoustical effect of the pinna is based upon reflection, shadowing, dispersion, diffraction, interference, and resonance" (Blauert, 1982, p. 63).

Despite the fact that HRTF's are highly complex and difficult to relate to a simple model, there is a simple perspective that might aid in our understanding of the overall behavior of HRTF's. The perspective focuses upon capturing directionally-dependent changes in the overall shape of HRTF's through coarse spectral analysis. Besides the narrow, deep notches that seem to migrate in frequency as a sound source changes direction (Butler & Belendiuk, 1977), there are broad spectral peaks and notches that can be used to describe the shape of the transfer functions associated with particular incidence angles. In fact, Hiranaka & Yamasaki (1984) have made similar distinctions between spectral fine structure and global spectral shape in their attempts to develop a simplified representation of the pattern of spectral changes associated with changing direction. If directional information is carried by large-scale shifts in overall spectral energy distribution, as Blauert (1982) has claimed, then perhaps an examination of such global behavior of horizontal plane HRTF's may shed some light on how broad spectral cues encode azimuth angle. Identifying these cues in the stimulus for directional hearing may aid in determining how the auditory system evaluates the complex spectral profile presented to the two ears.

Principal components analysis (PCA) is a technique which is well suited to the task of quantifying directionally-dependent trends in broad spectral variation in HRTF's. Similar applications of PCA to vowel spectra have yielded good results that can be used for vowel identification (e.g., Plomp, Pols, & van de Geer, 1967). The principal components obtained in such studies are regarded as "spectral shape factors," in contrast with the more conventional formant descriptions of speech spectra. Zahorian & Rothenberg (1981) point out that such PCA results best describe the overall shape of the speech spectra, and place less emphasis on the major spectral peaks. The same perspective is appropriate in the present investigation. Though our observations are made over direction rather than over collections of steady state vowel spectra, band energies should be regarded as variates capturing overall spectral shape just as in the speech studies. In this investi-

gation, broad spectral variation was measured via simulated critical band analyses of measured HRTF's for two subjects. Applying PCA fulfills the following two objectives:

- 1) The variation in HRTF spectral energy distribution across 24 spectral bands is reduced to a small number of principal components of variation that should support meaningful directional distinctions between HRTF's.
- 2) Spectral band weighting functions are determined for each of these components to define the contrasts in spectral energy distribution that account for the directional distinctions made by the HRTF's.

Once these objectives are achieved, each HRTF will be identified by a distinct set of scores on the small number of components. PCA also allows for the construction of novel HRTF spectral energy distributions through the inverse application of the weighting functions to modified data in the low-dimensional space defined by the principal components. Thus PCA has the desirable feature of allowing both analysis and resynthesis of directional variation in HRTF's.

2. Head-related transfer function (HRTF) measurement

Subjects were seated on a platform that could be rotated through 360 degrees in 10 degree steps. HRTF complex frequency responses were determined at 36 sound-source directions through Time-Delay Spectrometry (TDS) using the Crown TEF-10 analyzer. The TEF generated a sine-wave sweep that was delivered from an MDM TA-2 loudspeaker at a fixed location 2 meters from the entrance to the ear canal of the subject. A waxy cotton plug (Flent's brand ear-stopple) was placed into the subject's ear canal and molded flush to the ear-canal entrance. A Knowles BT-1759 microphone was embedded in the wax. The directionally-dependent component of the system response was isolated by dividing out a reference measurement from the measurements obtained with the microphone placed in the subject's ear canal. The reference measurement was made by placing the wax-embedded microphone at the end of a narrow rod pointed at the loudspeaker and located at the subjects ear level.

Data were obtained at 399 frequencies from 45 Hz to 18000 Hz on the TEF, and then digitally transferred to a Sun-2 computer. The data at DC were constructed by setting the real component to the magnitude response at 45 Hz and setting the imaginary component to zero (the DC point being necessary for subsequent analysis). Impulse responses were obtained through the application of a discrete inverse Fourier transform. The impulse responses were truncated to 256 points (7.11 msec at 36000 Hz) in preparation for critical band transformation.

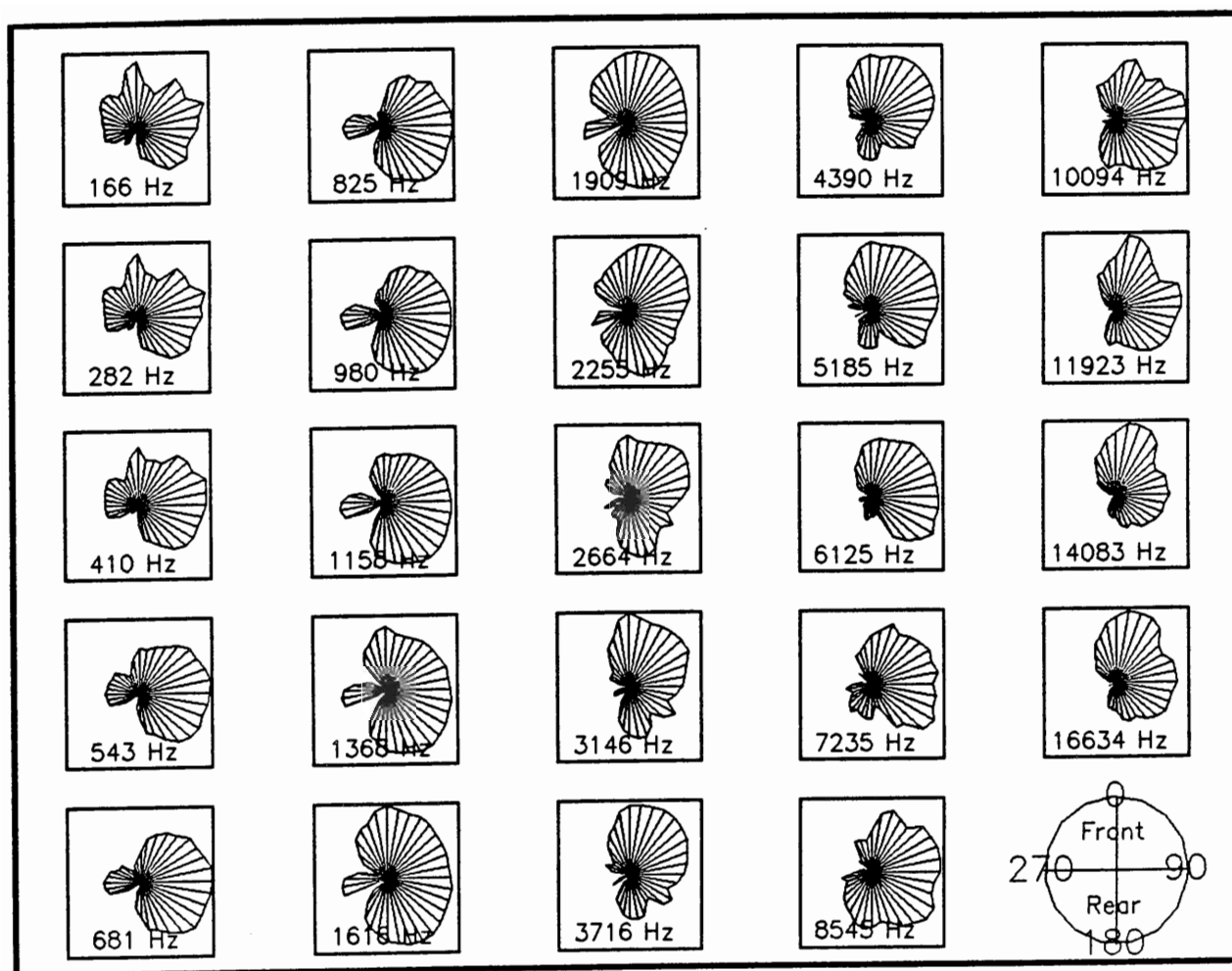


Figure 1. Rescaled HRTF log energy plotted over azimuth angle within each of 24 spectral bands for subject MDL. Peak frequency of each band appears in each polar plot.

3. Critical Band Analysis

In preparation for multivariate analysis, the HRTF data were processed according to a critical band transformation designed to model stimulus transduction in the peripheral auditory system. The premise behind this step was to base statistical analysis upon information as it would be available for human perception, in a domain similar to that in which perceptual analysis takes place. Impulse responses at 36 azimuth angles were processed by a bank of 24 FIR digital filters intended to model critical bands. The output of these filters provides an estimate of the energy that would be present in each critical band in response to a sonic impulse arriving from each of 36 directions. The critical band filter characteristics were based upon empirical measurements, both psychophysical and physiological, as described in Petersen (1980). As a precursor to PCA, the estimated critical band energies were log-transformed to simulate loudness transduction. Not

only is log-transformation generally regarded as a more perceptually relevant coding than linear coding (Stautner, 1983), but also in direct comparisons, Zahorian & Rothenberg (1981) showed that such coding of spectral band amplitude for PCA of speech spectra produced superior results when the intelligibility of resynthesized speech was used as a criterion.

The energy in each of the 24 spectral bands is illustrated as a function of azimuth angle in the polar coordinate plots of Figure 1. The legend in the lower right-hand corner of the figure shows how azimuth angle is organized in the polar plots: 0 degrees (frontward direction) is at the top of the plots and 180 degrees (rearward direction) is at the bottom of the plots. Adhering to Blauert's (1982) convention, ipsilateral directions are labeled 0 to 180 (the sound source and the measured ear are on the same side of the head) and contralateral directions are labeled 180 to 360 (the sound source is located on the side of the head opposite

the measured ear). For optimal graphical evaluation of each plot, band energy was rescaled from zero to one so that the direction in which band energy is at a minimum was plotted with a point at the origin. The panel in the upper right shows the energy in the lowest frequency band peaking at 166 Hz. Notice that the minimum energy for this band is in the contralateral-rear quadrant, in contrast to the 7235 Hz band within which energy is at a minimum in the contralateral-front quadrant. Plots for the second subject, DJF, showed the same pattern. Features such as these may be important in the perceptual decoding of spectral cues to direction. At present, it suffices to say that the application of PCA to these data is intended to identify such spectral features as potentially carrying directional information.

In further preparation for multivariate analysis, a decision was also made to subtract out differences in the mean level of the transformed HRTF's in the interest of examining only spectral differences between the horizontal plane transfer functions. Following this centering operation, each value was divided by the standard deviation of log energy at each direction in order to standardize the scaling of band energies for each HRTF. The result was used to construct a data matrix of 36 azimuth angles (rows) by 24 spectral bands (columns) with its rows exhibiting mean of zero and standard deviation of one. Though "row-centering" band energy in this way eliminates some of the information present in the raw band energies, it ensures that the PCA to follow will examine only spectral variation in HRTF's (in effect, it excludes directionally-dependent level differences in HRTF's from the analysis).

4. Principal Components Analysis

The data matrix $\mathbf{X}$ describing variation in band energy over azimuth may be viewed as a 24-dimensional representation of HRTF behavior on the horizontal plane. Each HRTF is associated with a single point in that 24-dimensional space. PCA was employed to transform the matrix $\mathbf{X}$ of 24 correlated variates into another matrix $\mathbf{Y}$ of 24 uncorrelated hypothetical variates that decrease in variance from first to last. The most intuitive explanation of PCA is that it rotates the axes of the 24-dimensional space until it finds the single vector along which variance is at a maximum. This vector defines the variate for the first column of $\mathbf{Y}$. PCA then searches for an orthogonal vector along which the residual variance is at a maximum, and continues searching until 24 orthogonal vectors account for all the variance in the data. The transformation determined through PCA is achieved by postmultiplying $\mathbf{X}$ by an orthogonal matrix $\mathbf{U}$, giving

$$\mathbf{Y} = \mathbf{X}\mathbf{U}$$

where $\mathbf{U}$ is a 24 by 24 matrix of weights on the variates of $\mathbf{X}$.

It should be noted that PCA is typically performed on data matrices that have been "centered by columns," and that the data matrix currently under analysis has been "centered by rows." Thus the variates do not express band energies relative to the mean log energy within each band, but rather the mean log energy within each HRTF. As compared with column-centering, double-centering (both rows and columns), and no-centering, the best agreement between the two subjects' PCA results (MDL and DJF) was obtained by row-centering. Coupled with the rationale that overall HRTF level differences should be eliminated from the data in order to isolate spectral differences, this finding led to the conclusion that row-centering was the best choice of the treatments.

In order to determine how many principal components should be included in the data reduction, the cumulative proportion of variance contained within the first several components was examined (the variances of the columns of $\mathbf{Y}$ that resulted from the rotation on $\mathbf{X}$ for two subjects). Adopting an arbitrary cutoff of 90 % selects four components for both subjects. Taken together, the first four components account for 91.3 % of the total variance for subject MDL, and 91.4 % of the variance for subject DJF. A more stringent criterion is also met in the comparison of the subjects' rotation weights, as described below.

Each component is defined by a vector of weights on the variates of $\mathbf{X}$, each vector forming a column of $\mathbf{U}$. The rotation weights defining the first principal component for subject MDL are illustrated in the upper right panel of Figure 2. These vectors of weights may be viewed as weighting functions that emphasize information in some critical bands (either positively or negatively), and de-emphasize information in others. Note that principal component 1 for both subjects weights low-frequency bands positively and high-frequency bands negatively. This suggests that the principal source of variation in centered horizontal plane HRTF's is a shifting of energy to and from high-frequency bands as azimuth angle varies. Such a shifting has classically been attributed to the acoustical shadowing effect of the head. Indeed, the high-frequency attenuation at the contralateral ear is so easily observed that one must expect it to be a prominent factor in HRTF behavior.

The second principal component obtained for each subject also matches quite well (lower right panel of Figure 2). While a comparison of the rotation weights defining the first principal component for each subject reveals a correlation coefficient of $r = 0.96$, the second principal components for the two subjects correlate at $r = 0.88$. It is clear that components 1 and 2 match across subject without confusion, but components 3 and 4 are less well matched between the two subjects. Although components 3 and 4 are strictly orthogonal for a single subject, they are fairly well correlated between subjects. For example, there is insufficient distinction between inter-subject correlations for DJF component 4 to allow a confident matching with MDL component 3 or 4 (correlations being $r = -0.57$ and

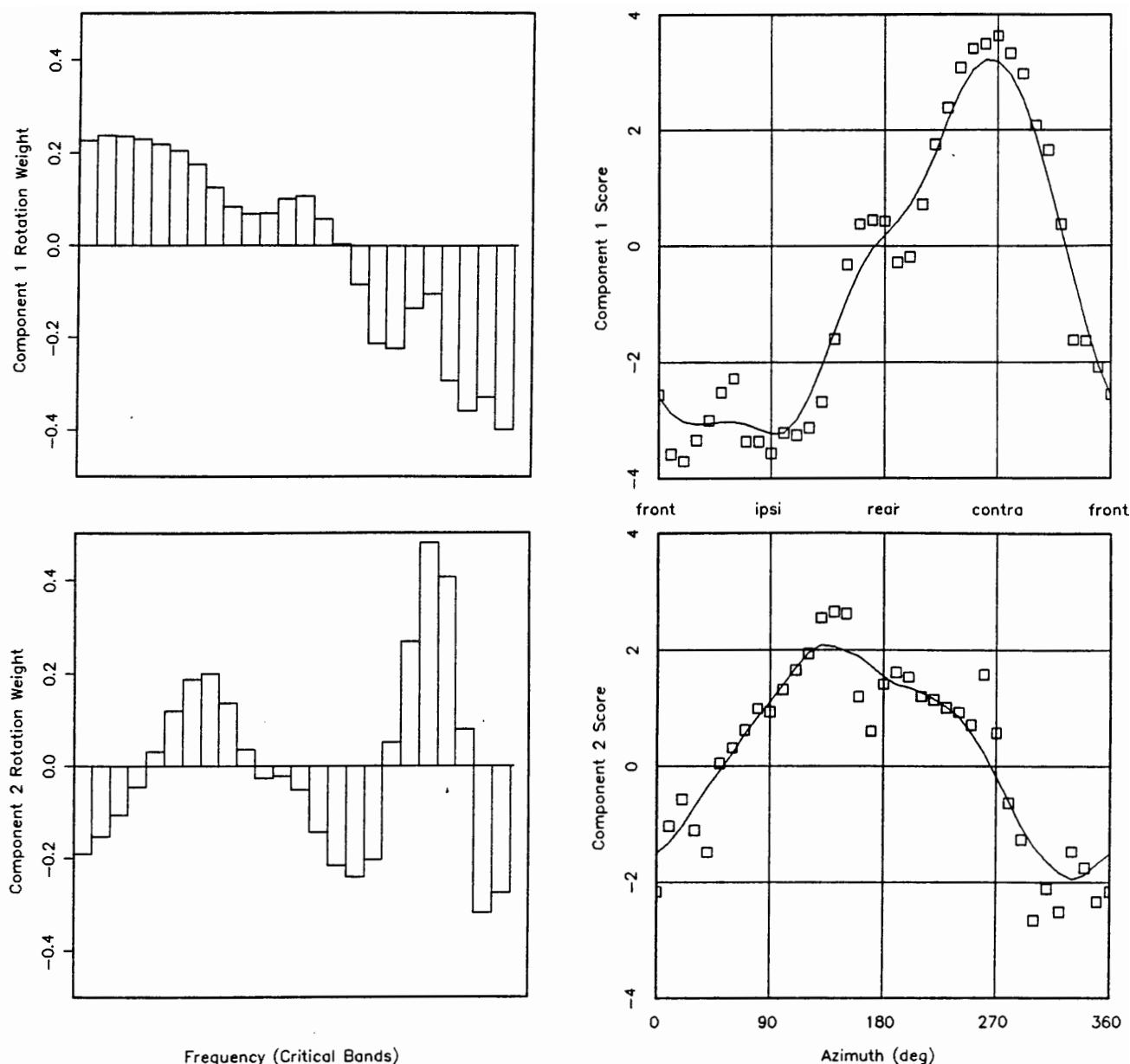


Figure 2. Rotation weights (right column) and scores (left column) on principal components 1 and 2 (rows) obtained for subject MDL (see text for details).

$r = -0.65$, respectively). The difference in the sign of the rotation weights is not important, as it defines the same distinction in band energies for the two subjects. Furthermore the weighting functions obtained for components 3 and 4 are difficult to interpret with respect to known characteristics of HRTF behavior, and so conclusions about these weighting functions must be stated in terms of the directional distinctions they seem to support (discussed below in section 5 on PC scores).

The rotation weights obtained for principal component 2 are interpretable with respect to known characteristics of HRTF behavior. The weighting functions defining this component divide the bands into four groups (frequency regions) for both subjects:

- 1) below band 10 (below 1.7 kHz)
- 2) band 11 to band 17 (from 1.7 to 5.5 kHz)
- 3) band 17 to band 21 or 22 (from 5.5 to 14 kHz)
- 4) above band 21 or 22 (above 14 kHz)

The frequency regions revealed by this principal component correspond closely to Blauert's (1982) "directional bands" that distinguish frontal directions from directions to the rear of the listener. Both from an analysis of HRTF's and from psychophysical investigations, Blauert concluded that perceptual decoding of azimuth could be based upon an assessment of the relative energy levels in a few broad directional bands (frequency regions) that he labeled either "v" for *vorne* (front) or "h" for *hinten* (rear). PCA on HRTF's for 36 azimuth angles supports his conclusion. Blauert's 4 to 6 kHz v directional band overlaps mostly with group 2 of bands described above. Blauert's 14 to 15 kHz v directional band coincides with group 4 of bands. Both groups 2 and 4 of bands received a positive rotation weight on this principal component. The 8.5 to 14 kHz h directional band is contained within group 3 of bands that receive a negative weight. The 1 to 2 kHz h directional band corresponds to group 1 of bands that also receive a negative weight. Scores on this principal component should distinguish well between HRTF's for front and rear directions. This prediction will be evaluated in the next section.

5. Principal Component Scores

As stated above in Section 3, principal component scores are calculated according to the matrix equation

$$Y = XU.$$

So the score for a given direction on a given component is a weighted sum of estimated critical band energies. For example, the score for the first azimuth (0 degrees) on the first principal component is:

$$y_{1,1} = x_{1,1}u_{1,1} + x_{1,2}u_{2,1} + \dots + x_{1,24}u_{24,1}$$

The upper left panel of Figure 3 plots scores on component 1 as a function of azimuth angle for subject MDL. The square symbols are individual scores and the smooth curve is the result of "LOWESS" locally weighted regression analysis (Cleveland, 1979). The portion of the data used in generating the smooth curve always spanned 120 degrees, and the smoothing at extreme azimuth values employed the appropriate data available through wrapping azimuth back onto itself at 360 degrees. As expected, component 1 distinguishes between ipsilateral and contralateral HRTF's quite successfully. For both subjects, the scores are predominantly negative from 0 to 180 degrees, and predominantly positive from 180 to 360 degrees.

The lower left panel of Figure 3 plots scores obtained on principal component 2. The smooth curve clearly shows that this component could support front-rear discrimination. The scores are predominantly negative from 90 to 270 degrees, and predominantly positive from 0 to 90 and 270 to 360 degrees. Taken together, the pair of components effectively discriminate the directions encoded by horizontal plane HRTF's. In effect, scores on the two components unambiguously

identify unique azimuth angles, even though a single score on one component can only indicate that the azimuth angle has one of two possible values.

Though component 3 was difficult to interpret strictly on the basis of the obtained weighting functions, an examination of the component scores resulting from this rotation revealed an interpretable distinction. While the obtained scores are virtually all negative, the scores reach minima (in absolute magnitude) at around 90 and 270 degrees and reach maxima at around 0 and 180 degrees. Thus, this component effectively distinguishes HRTF's of central versus lateral directions.

As with component 3, component 4 was difficult to interpret on the basis of obtained weighting functions. But again, an examination of the component scores revealed an interpretable distinction. MDL component 4 scores show minima at around 135 and 315 degrees and reach maxima at around 45 and 225 degrees. If the four quadrants of azimuth are labeled with numbers increasing as azimuth increases, starting with the ipsilateral-front quadrant as number 1, then component 4 distinguishes odd quadrants from even quadrants. Then MDL scores reach maxima in odd quadrants of azimuth angles (ipsilateral-front and contralateral-rear) while reaching minima in the even quadrants (ipsilateral-rear and contralateral-front).

6. Idealized principal component scores

If the four hypothetical variates found by PCA identify meaningful components of HRTF variation on the horizontal plane, then it could be that the perception of azimuth depends upon those components. There is sufficient psychophysical evidence (Blauert, 1982) to suggest that the first two principal components are relevant to directional hearing, but the weighting functions for components 3 and 4 have no clear precedent as carriers of directional information. Nonetheless, the four components account for over 90 % of the variance in the analyzed horizontal plane HRTF data, and the distinctions they make clearly are interpretable. Using smoothed principal component scores to reconstruct potentially superior HRTF's is one step in the direction of idealizing HRTF's. But if the interpretations given to the four components resulting from PCA are valid, then there is another step that could take us even closer to hypothetically ideal HRTF's. We can synthesize principal component scores that exhibit ideal behavior in the four-dimensional (4D) component space.

Four orthogonal sinusoidal curves were chosen to define ideal principal component scores. The frequency and phase of the sinusoidal curves were fit by eye to match directional interpretations of the obtained components. If we take F to be one cycle of variation over 360 degrees of azimuth, the chosen curves are described by sine and cosine functions at F and 2F. The amplitude of each sinusoidal curve was determined by the absolute maximum score observed on each component resulting from PCA on a matrix of data containing the averaged HRTF magnitudes of subjects MDL and DJF.

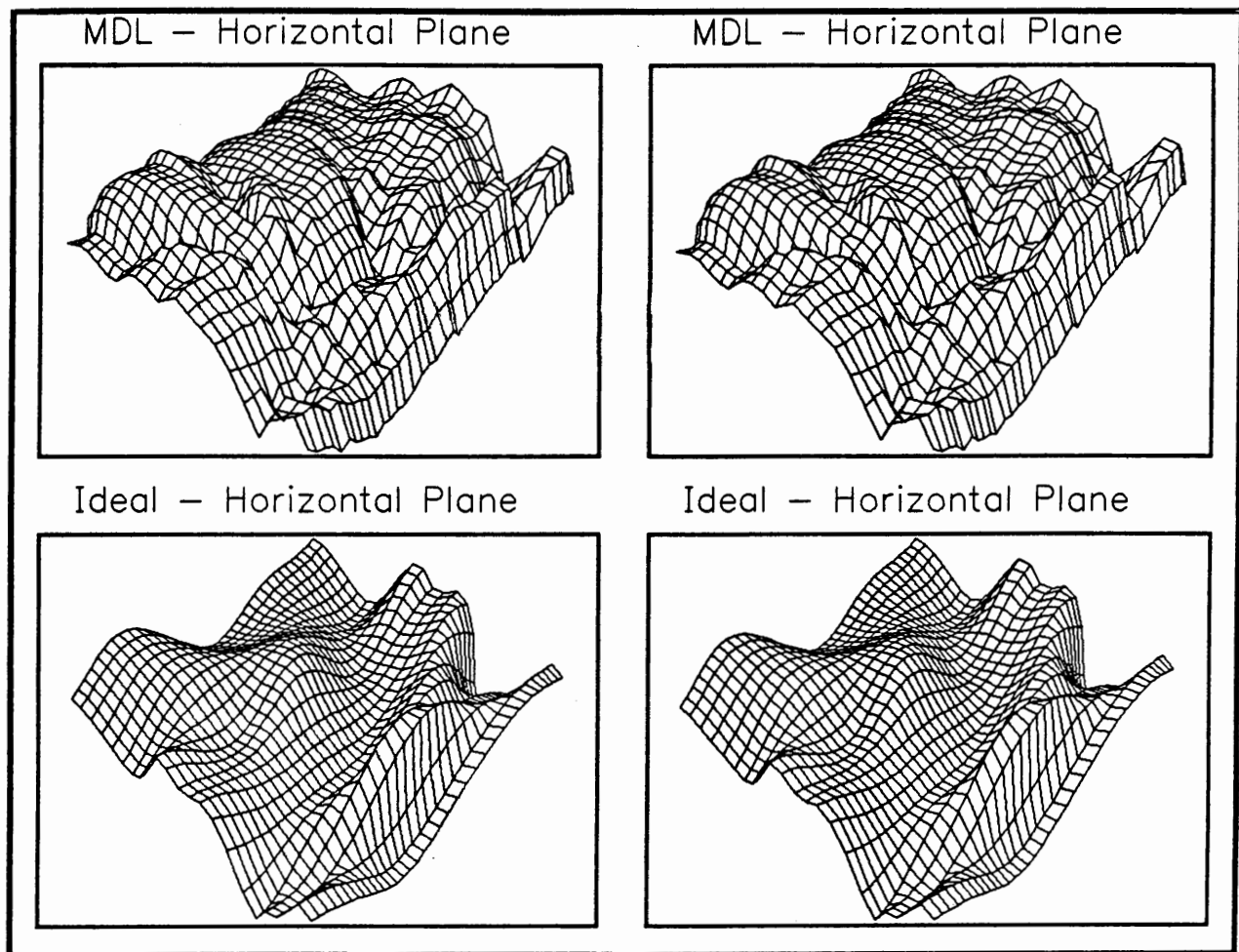


Figure 3. Stereographic perspective plots of azimuthal variation in MDL's transformed HRTF magnitude responses as measured (upper panels) and resynthesized (lower panels). Adjacent pairs of plots are intended for "free-fusion" to create a binocular depth percept. This fusion of the pair of plots is accomplished by crossing your eyes until the two are brought into alignment and then refocusing on the images. Some viewers have difficulty with this "free-fusion" due to the imbalance between convergence and accommodation that must be tolerated during viewing.

7. Resynthesis of HRTF's from idealized scores.

When Kramer & Mathews (1956) first suggested that a speech signal could be efficiently transmitted through "a linear coding" of a set of frequency-band signals, they placed emphasis upon resynthesizing the speech signal from as few transmission channels as possible. For the current PCA application, we will regard the HRTF as a directional signal for which an efficient transmission is desired. The preceding section showed that the 24-dimensional representation of HRTF behavior on the horizontal plane could be reduced to a phenomenon exhibiting only four interpretable dimensions of variation. Just as four component scores should identify an HRTF that has been transformed according to the principal components rotation matrix $\mathbf{U}$, so could an HRTF be practically reconstructed

from scores on those four components found in matrix $\mathbf{Y}$. If the small proportion of residual variance on the remaining 20 principal components is regarded as inconsequential, then the scores on components 5 through 24 can be set to zero to create a truncated matrix $\mathbf{Y}'$ of principal component scores. This type of reconstruction of the HRTF's for all 36 directions requires that we decode four sets of scores in the principal components domain through a reverse transformation back to the domain of the analyzed data. A new matrix $\mathbf{X}'$ of HRTF band energies can be constructed from the truncated matrix $\mathbf{Y}'$ of scores through postmultiplying by the inverse of the principal components rotation matrix:

$$\mathbf{X}' = \mathbf{Y}' \mathbf{U}^{-1}$$

It was hypothesized that using the idealized scores described in the previous section for Y^* would result in the construction of a matrix of HRTF's that contain more robust directional cues than the HRTF's measured for either subject. The implicit assumption here is that HRTF behavior should exhibit smooth spectral variation along dimensions relevant to directional perception. The distribution of obtained scores around the smooth curves would then be attributed to natural deviation of HRTF's from the ideal behavior that would give the best directional cues. Such an assumption is consistent with Butler & Belendiuk's (1977) observation that the pinnae supporting the best directional hearing show the smoothest migration of spectral features with changing azimuth angle.

A matrix of principal component scores was constructed from the idealized scores for the first four principal components. The remaining 20 columns of scores were set to zero. This matrix was postmultiplied by the inverse rotation matrix from the PCA results for MDL. In order to compare measured HRTF's and these resynthesized HRTF's two stereographic perspective plots were constructed. These plots illustrate the global pattern of HRTF variation for MDL's original and resynthesized horizontal plane HRTF's. They depict azimuthal variation in HRTF spectral energy distribution as smooth surfaces to be viewed stereoscopically. The surfaces were generated from the log-transformed, row-centered data by constructing a grid of datapoints connected by straight line segments over the plane defined by band number and azimuth angle. The upper panels of Figure 3 show the stereo pair of perspective plots over 360 degrees of azimuthal variation. The azimuth angle for each plot increases from the rear right edge of the plot to the front left edge of the plot. Therefore, the rightmost point is the datapoint for the lowest frequency band in the HRTF for 0 degrees azimuth. The leftmost point is constructed from the same datapoint, since the data wrap around the entire 360 degrees of azimuth. The corner of the plot that protrudes out of the page shows the datapoint for the highest frequency band at the same angle.

The question of whether the resynthesized HRTF's actually provide superior directional cues can only be answered through perceptual evaluation (to be undertaken in the confirmatory second stage of our directional hearing research project). Informal evaluation accomplished by listening to synthetic stimuli moving in a circular path around the head suggests that the HRTF's reconstructed from smoothed scores provide cues to directional motion that are superior to those of either subject. A discussion of how to collect perceptual data that might confirm or deny the superiority of one set of HRTF's over another is beyond the scope of this paper.

8. Conclusion

Principal components analysis reveals four aspects of the behavior of HRTF's on the horizontal plane. Variation in HRTF spectral energy distribution with chang-

ing azimuth angle as observed within 24 spectral bands can be captured by four principal components quantified in terms of spectral band weighting functions. These weighting functions can be used to evaluate synthetic directional transfer functions (DTF's) intended to simulate spectral cues to directional hearing on the horizontal plane. In effect, they can be used to determine whether the DTF's exhibit the same kind of overall variation in spectral energy distribution found for measured HRTF's. In the design of new sets of DTF's, ideal principal component scores can be used to generate hypothetically ideal spectral energy distributions over critical bands. Evaluation through principal components analysis is especially valuable in the context of automated DTF design driven from information in multiple domains of description, since it provides a simple (4D) measure of global spectral variation which is otherwise difficult to quantify.

9. References

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